



**Ministry of Higher Education and Scientific Research
University of Basrah - College of Engineering
Department of Computer Engineering**



SMART DIABETES MONITORING AND PREDICTION SYSTEM

A PROJECT

**Submitted to the Council of the Engineering College, University of Basrah,
in partial fulfilment of the requirements for the award of the degree of
Bachelor of Computer Engineering.**

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DECLARATION

We hereby declare that this project report entitled “**Smart Diabetes Monitoring and Prediction System**” is based on our own original work, except where references and citations have been duly acknowledged. This work has not been submitted, in whole or in part, for the award of any other degree or qualification at any university or institution.

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APPROVAL FOR SUBMISSION

I certify that this project report entitled "**SMART DIABETES MONITORING AND PREDICTION SYSTEM**", prepared by Ahmad Azhar Almansoor and Ibrahim Madian Fadhil, has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Computer Engineering at the University of Basrah.

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Supervisor: _____.

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Date: _____.

DEDICATION

To every soul bearing the silent weight of diabetes, the child who learned too young what it means to be different, the parent who smiles through the fear, the elderly who face each morning with needles and hope.

To those who were never warned, and to those who were warned too late. To the millions who's suffering the world has yet to fully see.

This work was not written in ink alone, it was written in the deep conviction that no life should be lost to a disease that science can predict, and technology can reach.

May every algorithm within these pages carry the weight of your dignity, and may the day come, soon, when early detection spares another family from grief.

This is for you.

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To those we hold dearest in our hearts, the ones whose presence made the difficult days bearable and whose memory made the quiet nights meaningful. You were never far from our thoughts throughout this journey, and this work carries a piece of every one of you within it.

Above all, we owe everything to our families, whose patience, sacrifice, and unconditional love carried us through every challenge. This achievement belongs to them as much as it does to us.

SMART DIABETES MONITORING AND PREDICTION SYSTEM

ABSTRACT

Diabetes mellitus affects an estimated 537 million adults globally, with approximately half of all cases remaining undiagnosed until irreversible complications arise. Conventional laboratory-based screening is insufficient to close this gap at population scale, creating a clinical imperative for accessible, data-driven early detection systems that operate without requiring clinical infrastructure.

This study presents a complete and reproducible machine learning pipeline for diabetes risk prediction applied to the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset, comprising 253,680 self-reported behavioral and demographic records. Seven classification algorithms were evaluated across three progressive experimental phases covering the original 21-feature representation, a domain-engineered 45-feature space incorporating WHO-standard BMI encodings, composite cardiovascular risk scores, pairwise interaction terms, and socioeconomic proxies, and a reduced top-15 feature subset selected through combined Random Forest Mean Decrease in Impurity and Mutual Information ranking. A 6.18:1 class imbalance was addressed through algorithm-level cost-sensitive weighting and `scale_pos_weight` parameterization. Model selection was governed by a clinically weighted composite scoring function assigning 40 percent weight to Recall, reflecting the asymmetric clinical cost of missed diabetic diagnoses. The best-performing model underwent systematic hyperparameter optimization via `RandomizedSearchCV` with Stratified 5-Fold Cross-Validation and Recall as the optimization criterion, followed by principled decision threshold optimization enforcing a binding clinical deployment constraint of $\text{Recall} \geq 0.90$.

LightGBM trained on the 45-feature engineered space achieved the highest composite score across all configurations. At the identified deployment threshold of 0.53, the final model correctly identifies 6,362 out of 7,019 diabetic cases in the held-out test set, achieving a Recall of 0.9064, ROC-AUC of 0.8186, PR-AUC of 0.4447, Balanced Accuracy of 0.7228, MCC of 0.3114, and F1-Score of 0.4066. The complete pipeline was deployed as a Streamlit web application providing a structured three-section interface, a landing panel, a 21-field patient information form, and a risk prediction output screen, enabling real-time risk assessment without requiring clinical infrastructure or programming expertise.

The results confirm that LightGBM with cost-sensitive configuration and clinically constrained threshold optimization constitutes a robust and deployable framework for population-level diabetes screening, with direct implications for preventive care delivery in resource-limited settings.

TABLE OF CONTENTS

PRELIMINARY PAGES

DECLARATION.....	I
APPROVAL FOR SUBMISSION.....	II
DEDICATION.....	III
ACKNOWLEDGEMENTS	IV
ABSTRACT.....	V
TABLE OF CONTENTS.....	VI
LIST OF FIGURES	XI
LIST OF TABLES	XII
LIST OF EQUATIONS	XIV
LIST OF ABBREVIATIONS	XV

CHAPTER ONE: INTRODUCTION

1. Introduction.....	1
1.1 Background and Motivation	1
1.1.1 Global Burden of Diabetes Mellitus	1
1.1.2 Economic and Social Impact.....	2
1.1.3 The Role of Artificial Intelligence in Healthcare	3
1.1.4 Need for Early Prediction Systems	3
1.2 Problem Statement.....	4
1.2.1 Limitations of Traditional Screening Methods	4
1.2.2 The Class Imbalance Challenge	5
1.2.3 The Cost of Missed Diagnoses	5
1.3 Research Objectives	6
1.4 Scope and Limitations of the Study.....	7
1.4.1 Scope.....	7
1.4.2 Limitations	8

1.5 Research Contributions	8
1.6 Report Organization	10

CHAPTER TWO: LITERATURE REVIEW

2. Literature Review	12
2.1 Overview of Diabetes and Prediction Systems	12
2.1.1 Types of Diabetes and Clinical Definitions	12
2.1.2 Risk Factors and Epidemiology	14
2.1.3 Evolution of Diabetes Prediction Systems	14
2.2 Machine Learning Techniques in Healthcare	15
2.2.1 Supervised Learning for Disease Classification	15
2.2.2 Ensemble Methods and Gradient Boosting	16
2.2.3 Handling Class Imbalance in Medical Datasets	17
2.2.4 Explainability and Interpretability in Clinical ML	18
2.3 Review of Previous Studies on Diabetes Prediction	18
2.3.1 Studies Using the Pima Indians Diabetes Dataset	19
2.3.2 Studies Using the BRFSS Dataset	19
2.3.3 Studies Using LightGBM and Gradient Boosting	20
2.3.4 Studies with Threshold Optimization	20
2.4 Comparative Analysis of Previous Works	21
2.5 Identified Research Gaps	23
2.6 Chapter Summary	24

CHAPTER THREE: METHODOLOGY

3. Methodology	25
3.1 System Overview	25
3.2 System Workflow and Design	25
3.2.1 Pipeline Architecture	26
3.2.2 Development Environment and Tools	27
3.3 Dataset Description	27

3.3.1 BRFSS 2015 Dataset Overview	27
3.3.2 Feature Descriptions and Data Types.....	27
3.3.3 Class Distribution and Imbalance Analysis	29
3.4 Data Preprocessing	30
3.4.1 Exploratory Data Analysis	30
3.4.2 Duplicate and Missing Value Treatment	30
3.4.3 Outlier Detection and Capping	31
3.4.4 Feature Scaling with RobustScaler	32
3.4.5 Imbalance Handling Strategy	32
3.5 Feature Engineering and Selection	33
3.5.1 Domain-Informed Feature Engineering	33
3.5.2 WHO BMI Category Encoding	33
3.5.3 Composite Risk Score Construction	33
3.5.4 Interaction Terms and Non-linear Features	33
3.5.5 Additional Engineered Features	34
3.5.6 Feature Importance Analysis and Top-15 Selection	37
3.6 Model Architecture	38
3.6.1 Baseline Models	38
3.6.2 LightGBM Architecture and Configuration	39
3.6.3 Imbalance Handling per Model	40
3.6.4 Composite Scoring for Model Selection	41
3.7 Training, Validation, and Testing Strategy	42
3.7.1 Train-Test Split Strategy	42
3.7.2 Stratified K-Fold Cross-Validation	42
3.7.3 Hyperparameter Tuning with RandomizedSearchCV	42
3.8 Evaluation Metrics	44
3.8.1 Recall and Precision	44
3.8.2 F1-Score and Balanced Accuracy	44
3.8.3 ROC-AUC and PR-AUC	45
3.8.4 Matthews Correlation Coefficient	45

3.8.5 Decision Threshold Optimization	45
3.9 Final Pipeline Bundle	47
3.10 System Deployment	47
3.11 Chapter Summary	50

CHAPTER FOUR: RESULTS AND DISCUSSION

4. Results And Discussion	51
4.1 Experimental Setup	51
4.2 Model Training Results	52
4.2.1 Baseline Phase Results	52
4.2.2 Engineered Features Phase Results	54
4.2.3 Selected Features Phase Results	56
4.2.4 Global Model Comparison	57
4.3 Performance Evaluation	59
4.3.1 Hyperparameter Tuning Results	59
4.3.2 Threshold Optimization Analysis	61
4.3.3 Confusion Matrix Analysis	62
4.3.4 ROC and Precision-Recall Curves	63
4.3.5 Learning Curve Analysis	65
4.3.6 Final Model Results at Deployment Threshold	66
4.4 Comparison with Existing Literature	67
4.5 Discussion and Analysis	69
4.5.1 Interpretation of Key Results	69
4.5.2 Clinical Significance of Recall Optimization	70
4.5.3 Feature Importance and Clinical Alignment	70
4.6 Chapter Summary	72

CHAPTER FIVE: CONCLUSION AND FUTURE WORK

5. Conclusion And Future Work	74
5.1 Summary of Findings and Contributions	74
5.2 Limitations	76
5.3 Future Work and Recommendations	77
5.3.1 External Validation on Independent Datasets	77
5.3.2 Integration of Biomarker Features	77
5.3.3 Advanced Imbalance Handling with ADASYN	77
5.3.4 Bayesian Hyperparameter Optimization	78
5.3.5 Longitudinal Data Integration	78
5.3.6 Formal Clinical Evaluation and Regulatory Pathway	78
5.4 Chapter Summary	79

RESEARCH SUMMARY

1. Objective of the Study	80
2. Research Methodology	80
3. Sample and Research Instrument	81
4. Key Findings of the Study	82
5. Research Implications and Key Recommendations	83

REFERENCES

References	84
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LIST OF FIGURES

Figure	Caption	Page
Figure 2.1	Global diabetes prevalence among adults aged 20–79 years, 2000–2050. Source: IDF Global Diabetes Report, 2024 [1]	13
Figure 2.2	Level-wise versus leaf-wise tree growth in LightGBM. Source: Ke et al., NeurIPS 2017 [15]	17
Figure 3.1	System architecture and pipeline workflow of the Smart Diabetes Monitoring and Prediction System	26
Figure 3.2	Point-biserial correlation coefficients between each original BRFSS 2015 feature and the Diabetes_binary target variable	29
Figure 3.3	Class distribution of the Diabetes_binary target variable in the BRFSS 2015 dataset showing the 86/14 majority-to-minority imbalance ratio	30
Figure 3.4	Boxplots of continuous features illustrating value ranges and outlier distribution prior to IQR-based capping, BRFSS 2015 dataset	31
Figure 3.5	Effect of SMOTE oversampling on class distribution before and after minority class augmentation on a 10,000-record training subset	32
Figure 3.6	Distribution of engineered features by diabetes class in the BRFSS 2015 dataset	36
Figure 3.7	Combined Random Forest MDI and Mutual Information consensus ranking of the top 15 selected features	37
Figure 3.8	Top 20 hyperparameter configurations evaluated during RandomizedSearchCV ranked by mean cross-validated Recall score with standard deviation error bars	43
Figure 3.9	Variation of Recall, Precision, F1-Score, and MCC across decision thresholds from 0.05 to 0.95 for the tuned LightGBM model with the clinical deployment threshold of 0.53 marked	46
Figure 3.10	Streamlit application landing section showing the system title, clinical description, and model performance panel.	48
Figure 3.11	Streamlit application patient information form showing the three-column input layout for all 21 BRFSS variables.	49
Figure 3.12	Streamlit application result screen for a low-risk classification showing the probability score, threshold bar, and clinical recommendation.	49

Figure 3.13	Streamlit application result screen for a high-risk classification showing a predicted probability of 97.2% and all identified risk factors.	50
Figure 4.1	Performance comparison of all seven algorithms across key metrics in Phase 1 (Baseline)	54
Figure 4.2	Comparison of key metrics between Phase 1 (original features) and Phase 2 (engineered features) per algorithm	55
Figure 4.3	Three-phase comparison of Recall, F1, Precision, and ROC-AUC per algorithm across all three feature phases	57
Figure 4.4	Global ROC-AUC heatmap showing performance of all model-phase combinations	58
Figure 4.5	Pre-tuning versus post-tuning comparison bar chart for LightGBM across key metrics	61
Figure 4.6	Confusion matrices at four threshold operating points: Default 0.50, F1-optimal 0.80, Youden J 0.661, and clinical deployment threshold 0.53	63
Figure 4.7	Final ROC curve and Precision-Recall curve for the tuned LightGBM model	64
Figure 4.8	Learning curve of the tuned LightGBM model showing training and cross-validated validation Recall as a function of training set size	65
Figure 4.9	Final confusion matrix of the tuned LightGBM model at deployment threshold 0.53	67
Figure 4.10	Feature importance ranking of the top 15 selected features for the tuned LightGBM model	72

LIST OF TABLES

Table	Caption	Page
Table 2.1	Methodological characteristics of selected prior studies on machine learning-based diabetes prediction	22
Table 3.1	Description, data type, and value range of all 21 original BRFSS 2015 features	28
Table 3.2	Complete list of 24 engineered features, construction formula, output type, and clinical rationale	34

Table 3.3	Configuration of the seven classification algorithms used across all experimental phases including imbalance handling mechanism	38
Table 3.4	Clinically weighted composite scoring function used for model selection across all experimental phases	41
Table 3.5	LightGBM hyperparameter search space used in RandomizedSearchCV with n_iter=50 and Stratified 5-Fold cross-validation	43
Table 4.1	Experimental setup summary	52
Table 4.2	Performance of all seven algorithms on the original 21 BRFSS features (Phase 1), sorted by composite score	53
Table 4.3	Performance of all seven algorithms on the 45-feature engineered space (Phase 2), sorted by composite score	55
Table 4.4	Performance of all seven algorithms on the top-15 selected features (Phase 3), sorted by composite score	56
Table 4.5	Global composite score comparison across all three experimental phases	57
Table 4.6	Optimal LightGBM hyperparameter values selected by RandomizedSearchCV	59
Table 4.7	Pre-tuning versus post-tuning performance comparison for LightGBM	60
Table 4.8	Comparison of four threshold operating points for the tuned LightGBM model	61
Table 4.9	Confusion matrix outcomes at clinical deployment threshold 0.53	62
Table 4.10	Final deployment performance metrics for the tuned LightGBM model at threshold 0.53	66
Table 4.11	Comparison of this study with representative prior literature on machine learning-based diabetes prediction	68
Table 4.12	Top 15 selected features, importance rank, and clinical significance	71

LIST OF EQUATIONS

Equation	Description	Page
Equation 3.1	IQR lower bound for outlier capping	31
Equation 3.2	IQR upper bound for outlier capping	31
Equation 3.3	Random Forest MDI feature importance	37
Equation 3.4	Mutual information between feature and target	37
Equation 3.5	Logistic regression sigmoid function	38
Equation 3.6	Binary Gini impurity	39
Equation 3.7	Gradient boosting additive model	39
Equation 3.8	Binary cross-entropy loss function	39
Equation 3.9	Balanced class weight formula	40
Equation 3.10	scale_pos_weight formula	40
Equation 3.11	Clinically weighted composite scoring function	41
Equation 3.12	Cross-validation mean performance estimate	42
Equation 3.13	Recall	44
Equation 3.14	Precision	44
Equation 3.15	F1-Score	44
Equation 3.16	Balanced accuracy	44
Equation 3.17	ROC-AUC probabilistic interpretation	45
Equation 3.18	Matthews correlation coefficient	45
Equation 3.19	Youden index threshold selection	46

LIST OF ABBREVIATIONS

Abbreviation	Full Term
ADASYN	Adaptive Synthetic Sampling
AI	Artificial Intelligence
AUC	Area Under the Curve
BalAcc	Balanced Accuracy
BMI	Body Mass Index
BRFSS	Behavioral Risk Factor Surveillance System
CDC	Centers for Disease Control and Prevention
CV	Cross-Validation
EDA	Exploratory Data Analysis
EHR	Electronic Health Record
F1	F1-Score (Harmonic Mean of Precision and Recall)
FN	False Negative
FP	False Positive
FPG	Fasting Plasma Glucose
FPR	False Positive Rate
GOSS	Gradient-based One-Side Sampling
HbA1c	Glycated Hemoglobin A1c
IDF	International Diabetes Federation
IQR	Interquartile Range
L1	L1 Regularization (Lasso)
L2	L2 Regularization (Ridge)
LightGBM	Light Gradient Boosting Machine
MCC	Matthews Correlation Coefficient
MDI	Mean Decrease in Impurity

MI	Mutual Information
ML	Machine Learning
NCD	Noncommunicable Disease
OGTT	Oral Glucose Tolerance Test
PR-AUC	Precision-Recall Area Under the Curve
RF	Random Forest
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-Sampling Technique
SVM	Support Vector Machine
T2DM	Type 2 Diabetes Mellitus
TN	True Negative
TP	True Positive
TPR	True Positive Rate
WHO	World Health Organization
XGBoost	Extreme Gradient Boosting